Natthabhat Nilsuwan (Namon) 6730154821@student.chula.ac.th / Tel.091-793-4343

SUMMARY

Third-year Mechanical Engineering student at Chulalongkorn University with hands-on experience in manufacturing, embedded systems, software development, and automotive engineering. Completed an internship at Western Digital, independently developing and deploying an AI-integrated machine inventory system. Founded NS Labs, a 3D printing business generating over 100,000 THB in six months, and contributed to CU Formula Student's second-place finish at TSAE Formula 2025. Seeking a 3–4-month engineering internship from May to August 2027.

EDUCATION

B.Eng. Mechanical Engineering, Chulalongkorn University **2024 – 2028(Expected)**

Availability: Full-time, 3-4 months internship from May to August 2027; flexible extension

Coursework: Statics, Dynamics, Thermodynamics, Fluid mechanics, Mech of Mat, Heat transfer, Design Elements

Interest: Embedded systems, Automotive, Motorbike, EV technology, IoT, Manufacturing, DIY, AI

EXPERIENCE (More at https://profile.natthabhat.com/)

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Entrepreneurship – Founder of a 3D Printing Business (First-year student, aged 19) 2

- Generated 100,000+ THB in revenue within 6 months as a solo founder.
- Designed custom parts using CAD and slicer software for local and global customers
- Specialized in FDM manufacturing, with exposure to SLA, SLS, and molding concepts
- Managed the full workflow: design, production, quality control, packaging, and logistics
- Built a website for online price estimates and sold products through global marketplaces.

Western Digital Internship – Maintenance Engineer (2 months) 3

- Independently developed and deployed a centralized machine inventory application.
- Supported standardized naming and established a centralized equipment database.
- Consolidated laboratory equipment records and implemented searchable profiles, status history, and access controls.
- Integrated an AI assistant and automated email notifications, then handed over the system for continued staff use.

CU Formula Student Team (Rapidamete) – Electrical Team Member 5

- 2nd place, TSAE Formula Student 2025; passed electrical inspection without issues
- Wired vehicle electrical systems and sensors
- Built a wireless timing gate to evaluate vehicle performance
- Developing an EV Formula car: HV battery, powertrain, and liquid cooling

Smart Home & IoT System (Independent Project) 6

- Integrated multi-brand IoT devices into a unified system
- Built and maintained Linux servers (remote access, media, AI, game hosting)
- Specialized in networking and self-hosted infrastructure

Programming and Hardware Projects 7

- Worked with ESP32, Arduino, and Raspberry Pi
- Built systems with wireless telemetry, user interfaces, stepper motors, and IR sensing
- Designed custom enclosures, wiring, and system architecture

CU Engineering Camps 8

- Planned and executed 4 engineering camps with 800+ participants
- Managed site mapping, participant routing, and on-site event coordination
- Developed interpersonal skills through collaboration

TECHNICAL SKILLS

Manufacturing: CAD Fusion 360, FDM and SLA 3D printing, CNC, laser cutting, material optimization

Embedded: ESP32, Arduino, Raspberry Pi, MQTT, Zigbee, BLE, Matter

Programming: Python, C, C++, HTML, CSS, JavaScript, React, PostgreSQL, Next.js, Node.js, Docker

Systems: Linux, networking, IoT, server deployment, and website hosting

Simulation: MATLAB, Simulink, Simscape, ANSYS

Languages: Thai (native) and English (B2) | CU-TEP : 92 / 120, CU-ATS : 1100 / 1600, TU-GET : 790 / 1000

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Entrepreneurship – Founder of a 3D Printing Business

Founded and independently operated **NS Labs**, a 3D printing business delivering both proprietary products and made-to-order manufacturing for domestic and international customers. The business specializes in FDM 3D printing, producing items ranging from small functional components to large-scale collectible models. The business generated 100,000 THB in revenue within 6 months through domestic and international online marketplaces. (2025 – present)

Business Development & Operations

- Planned and launched the business roadmap from initial product concept to market release
- Designed products using CAD software and optimized prints via slicer configuration
- Managed full production workflow: printing, quality control, post-processing, and packaging
- Conducted product photography, online listings, advertising, and customer communication
- Handled domestic and international logistics, shipping, and order fulfillment
- Scaled operations by reinvesting profits in additional 3D printers and the business website

Technical & Manufacturing Expertise

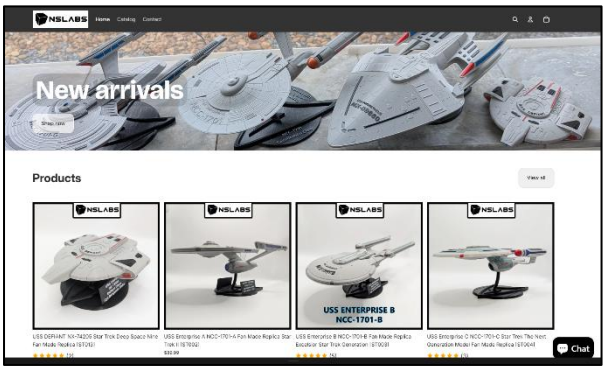
- Specialized in **FDM manufacturing** with practical exposure to SLA, SLS, and molding concepts
- Optimized material selection, print orientation, and process parameters for strength and quality
- Performed printer maintenance, troubleshooting, and reliability improvements

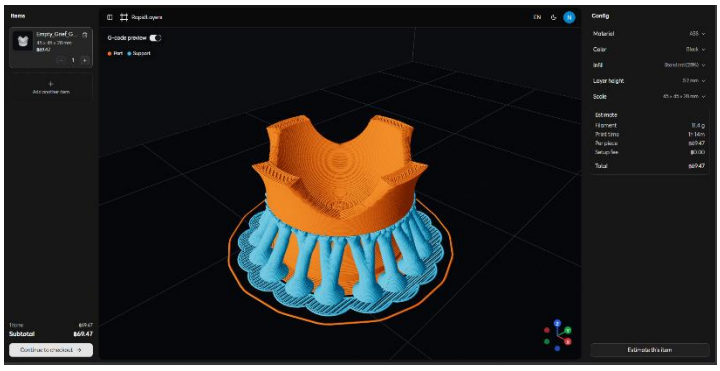
Problem Solving & Risk Management

- Resolved product quality issues through root-cause analysis and iterative design refinement
- Managed equipment downtime and failures through preventative maintenance
- Navigated international logistics challenges, platform fees, tariffs, and taxation
- Adapted pricing and workflow to mitigate financial loss and improve margins

Outcome

- Successfully stabilized business operations while balancing full-time university studies
- Gained hands-on experience in manufacturing systems, operations management, and applied engineering decision-making beyond classroom learning











Western Digital Internship – Maintenance Engineer (2 months)

Through the BOI Smart Manufacturing 5.0 Internship Program (2026), I completed over 100 hours of KU Mastery courses and workshops before undertaking a machine inventory monitoring project under the guidance of a mentor in the Maintenance Department. I identified staff pain points and independently developed a web application to centralize machine inventory records previously maintained in separate Excel spreadsheets.

Background

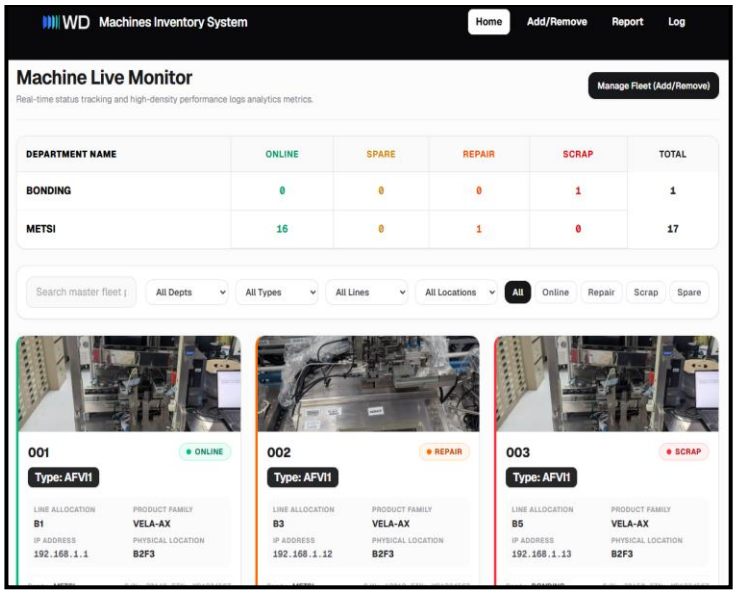
Before the project, laboratories and departments managed machine inventories using separate Microsoft Excel spreadsheets. This decentralized approach made assets difficult to locate and led to inconsistent equipment names, causing confusion when coordinating equipment across facilities.

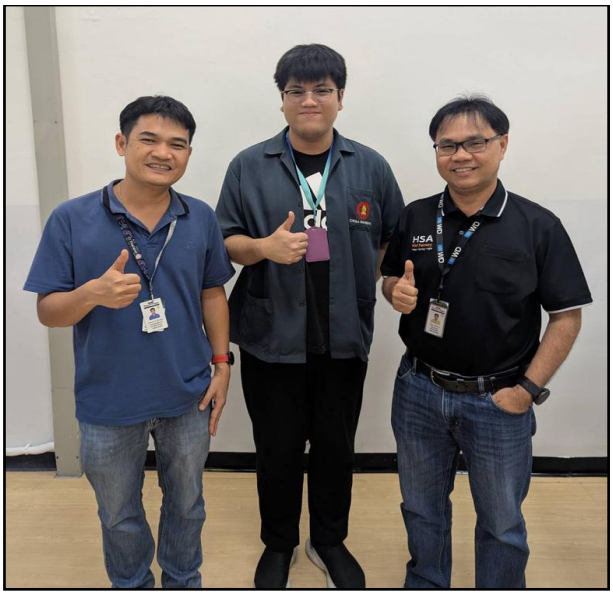
My Contribution

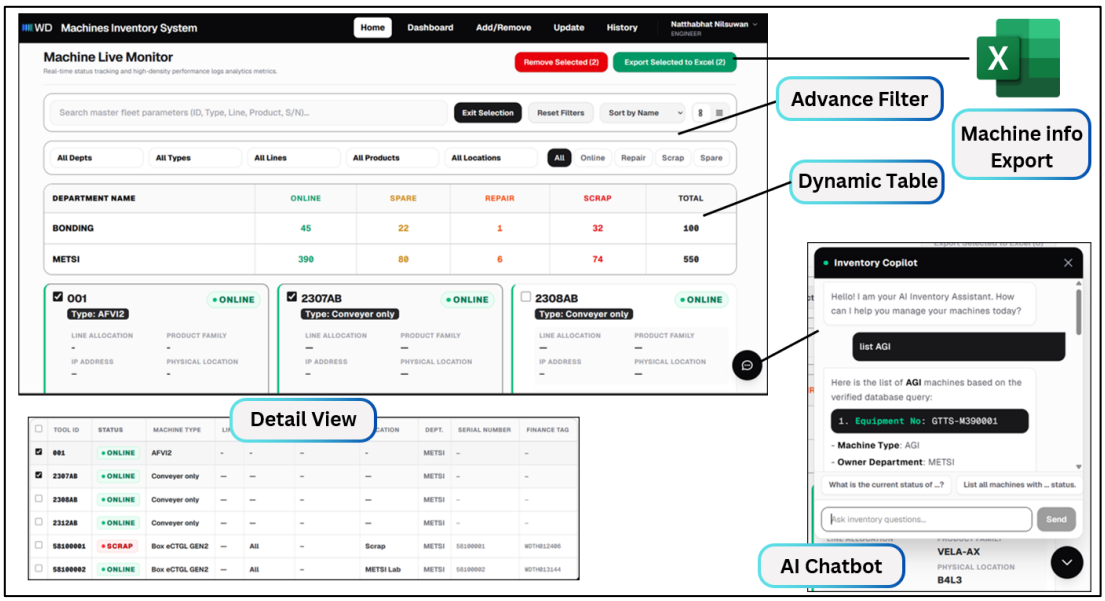
- Interviewed laboratory staff to identify inventory tracking problems and define system requirements.
- Developed the frontend and backend using Next.js and React, with a PostgreSQL database.
- Consolidated laboratory inventory data and supported standardized equipment naming.
- Implemented equipment search, status history, activity logs, Excel exports, and role-based access control.
- Integrated an AI assistant using Ollama with Typhoon 2.5 and Qwen 3.6 to query inventory information, and implemented automated daily email notifications.
- Deployed the application on WD servers, incorporated staff feedback, and handed over the source code and system to the mentor.

Outcome & Impact

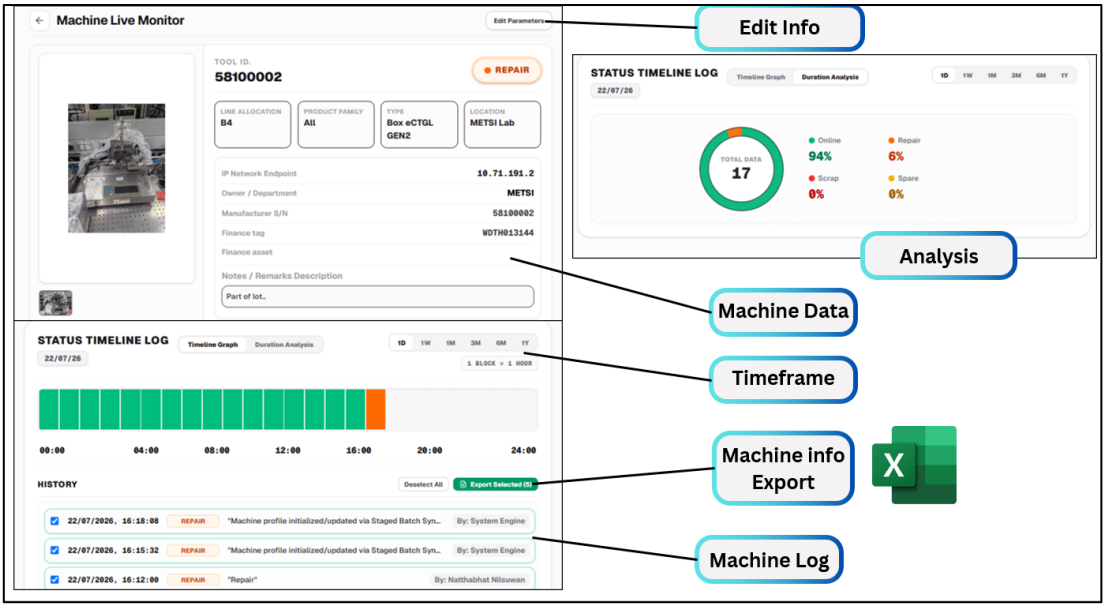
The system gives staff easier access to machine data, reduces confusion, and saves time when locating equipment information. It supports analysis of individual machines and maintains a history of their status changes and associated records. Through this project, I gained hands-on experience in enterprise database architecture, AI integration, and server deployment.



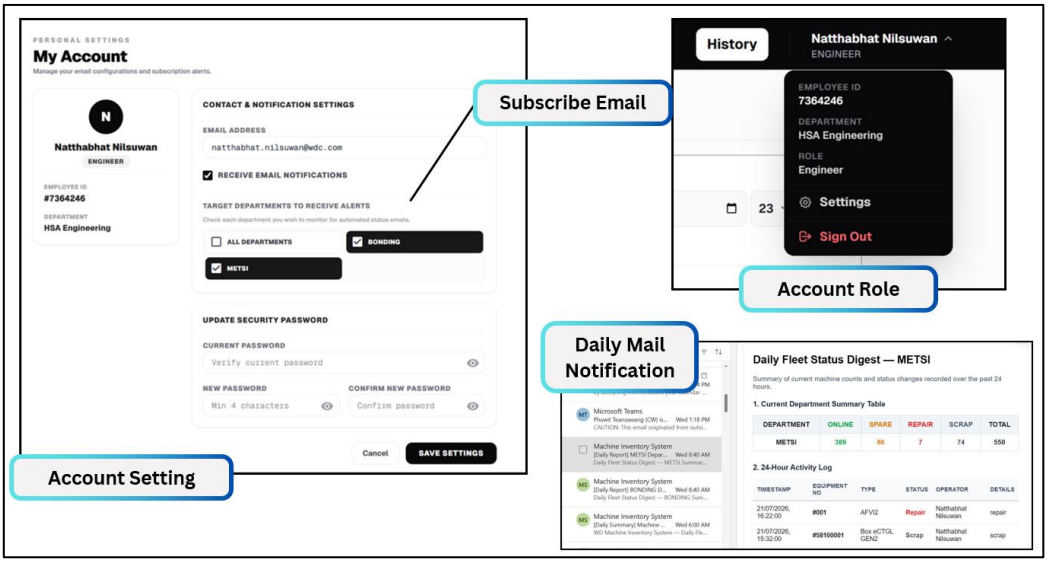




Home Page



Equipment Detail



CU Formula Student Team (Rapidamete) – Electrical Team Member

I joined the Formula Student team in my first year, driven by an interest in automotive and electric vehicle engineering. I chose the Electrical Team because of my embedded systems experience and the team's transition to an electric Formula Student car. The electrical team consisted of 8 members from first- to fourth-year students, working collaboratively to ensure system reliability and safety.

Responsibilities

- Designed complete vehicle wiring diagrams covering sensors, control units, and power distribution
- Completed the race car wiring, emphasizing reliability, safety, and secure connections
- Prepared cost reports in compliance with Formula Student regulations
- Assembled and tested the high-voltage battery system
- Researched EV motor and powertrain systems for performance and compatibility
- Collaborated with other teams to research and design an EV liquid cooling system

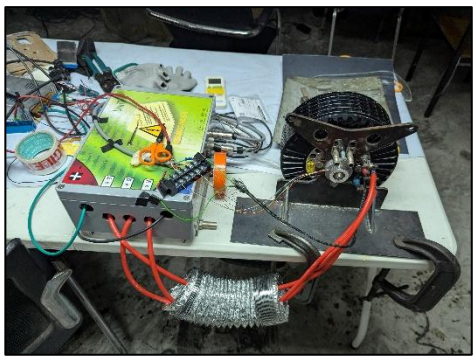
Technical Contribution

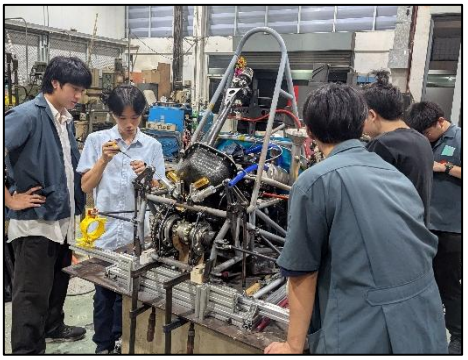
- Applied hands-on skills including soldering, wiring harness assembly, and electrical troubleshooting
- Interpreted and compared component datasheets to select appropriate parts
- Coordinated component sourcing and ordering directly from manufacturers
- Built a working wireless timing gate for vehicle testing
- Supported rapid prototyping and testing cycles under strict competition deadlines

Outcome & Impact

- Gained real-world engineering experience through practical tool usage and collaboration with experienced staff and senior students
- Strengthened engineering discipline, time management, and cross-team communication skills
- Built strong teamwork and professional relationships within a high-pressure project environment
- Contributed to the team's second-place finish at TSAE Formula Student 2025, competing against 16 universities nationwide







Smart Home & IoT System (Independent Project)

Designed and implemented a fully integrated smart home system, combining hardware, networking, and server-side software into a unified and scalable infrastructure. Researched and evaluated multiple IoT brands available in Thailand, comparing protocols, reliability, and ecosystem limitations to select devices that met the system requirements.

Hardware & IoT Implementation

- Installed and configured smart devices across all rooms, including:
  smart light switches, motion sensors, smart speakers, and environmental sensors
- Evaluated strengths and weaknesses of different manufacturers and communication protocols (Wi-Fi, Zigbee, BLE, Matter, Thread)
- Integrated devices from multiple brands into a single interoperable system

Software, Networking & Systems

- Learned and applied networking fundamentals, including LAN configuration and device segmentation
- Worked extensively with Linux operating systems for system administration
- Used Home Assistant to unify different IoT devices into one centralized control platform
- Developed foundational skills in web programming and Docker containerization

Server & Infrastructure Setup

Built a self-hosted server using a repurposed PC and managed the following services:

- **Media streaming:** Jellyfin, with media management using Seerr, Sonarr, and Radarr.
- **File sharing and photo management:** SMB/FTP file sharing, Immich, and PhotoPrism
- **Local AI services:** Ollama and llama.cpp running models such as Qwen, DeepSeek, Gemma, and Typhoon to support 3D printing workflows and smart home automation.
- **Website hosting:** Hosted a 3D printing storefront, printer automation tools, and a portfolio website using Cloudflare Tunnel.
- **Network infrastructure:** Organized server and networking equipment in a rack with UPS backup power to improve reliability. Used **Grafana** to visualize monitoring data and **netstat** to inspect network connections and listening ports.

Outcome

- Gained strong practical experience in hardware–software integration, networking, Linux systems, and IoT architecture
- Developed skills directly applicable to modern engineering environments
- Built a foundation that supports adaptation to future engineering coursework, team projects, and real-world system design







Programming and Hardware Projects

I developed an interest in computers and embedded systems from an early age, driven by curiosity about their capabilities and real-world applications. I researched platforms for experimentation and prototyping, beginning with Arduino microcontrollers and later expanding to ESP32, Raspberry Pi, and x86-based systems. Designed and built multiple projects integrating a wide range of peripheral devices, while learning to select appropriate platforms based on performance limits, power constraints, and system requirements.

Gained hands-on programming experience in C, C++, and Python, applying these skills across academic and independent projects. This foundation enabled effective participation in engineering activities inside and outside the classroom and confirmed a long-term passion for embedded systems and applied engineering.

Selected Projects

Randomness Generator (2021)

- Developed an interactive device using **Arduino Pico**, OLED display, and physical buttons
- Designed a graphical user interface that provided a foundation for future UI projects
- Implemented logic using Arduino randomization libraries to generate topic-based words

LINE Chatbot – Study Advisor (2022)

- Built a chatbot using **Google Dialogflow** to analyze academic performance
- Predicted learning outcomes and provided study recommendations
- Used accumulated user data to improve the chatbot through machine learning

Wireless Timing Gate System (2023)

- Designed a wireless timing system using **dual ESP32 devices** connected via **ESP-NOW**
- Achieved millisecond-level time synchronization using internet-based time references
- Developed an **IR sensing system resistant to sunlight interference**, suitable for outdoor use
- Optimized firmware to address overheating issues caused by high processing load

Zigbee Remote Controller (2024)

- Built a low-power remote controller using **ESP32**, multiple buttons, and a lithium battery
- Integrated the device with **Home Assistant** to control smart home devices
- Utilized **Zigbee protocol** for efficient power consumption and reliability

Remote-Controlled Construction Crane (2025)

- Designed and built a crane system from the ground up using **dual ESP32 devices**
- Controlled servo motors for both the crane and the remote unit
- Developed a user-friendly remote with an interactive OLED interface
- Designed the truss and chassis structures based on **statics and structural engineering principles**
- Integrated a lithium-ion battery system, achieving functionality comparable to commercial products

Programming and Hardware Projects (continued)







CU Engineering Camps

Participated as a **Planning Team member** in multiple large-scale engineering camps, including **Larngear 24, FE 22, Vishnu 24, and Larngear 25**. Contributed to the planning and execution of events involving **800+ participants**, ensuring smooth operations and minimal disruption throughout each program. (2025 - 2026)

Responsibilities

- Mapped and coordinated locations for teams and activity zones
- Designed and executed participant flow plans between activity points
- Guided participants efficiently from point A to point B
- Supported on-site operations to ensure events followed planned schedules
- Collaborated with multiple teams to resolve real-time logistical challenges

Skills & Outcomes

- Strengthened **communication, collaboration, and presentation skills**
- Developed **time management and people coordination** abilities in high-pressure environments
- Gained experience working with large, diverse groups inside and outside the faculty
- Contributed to delivering organized, safe, and engaging experiences for all participants





